1.0

$P(n) \coloneqq (\varphi(n{+}4) \geq \psi(n{+}4))$

SPf

BC

$$\begin{array}{lll} & \varphi(4) & \psi(4) \\ = & 4! & = 2^4 \\ = & 24 & = 16 \end{array}$$

$$\frac{24 \geq 16}{P(0)}$$

RcS

$$\frac{P(n)}{\begin{array}{ll} & \varphi(n{+}5) \\ = & (n{+}5)\varphi(n{+}4) \\ \geq & (n{+}5)\psi(n{+}4) \\ \geq & 2\psi(n{+}4) \\ = & \psi(n{+}5) \end{array}}{P(n{+}1)}$$

EPf

3.0

$a,b,c \in \mathbb{N}$

$\dfrac{a = 1a}{a|a}$

3.1

$\dfrac{\forall a \in \mathbb{N}\ (a|a)}{\text{Rflx}(|)}$

3.2

$\dfrac{a=0\ a|b}{b = c \bullet 0 = 0 = a}$

3.3

$\dfrac{a \neq 0\ a|b\ b|a}{b=ca\ a=db}$

3.4

$\dfrac{a = db = d(ca) = (dc)a}{dc = 1}$

3.5

$\dfrac{dc = 1}{d=1\ c=1}$

3.6

$\dfrac{d=1\ c=1}{b = ca = 1a = a}$

3.7

$\dfrac{\forall a,b \in \mathbb{N}((a|b\ b|a)/(a=b))}{\text{AntSym}(|)}$

3.8

$\dfrac{a|b\ b|c}{b=c'a\ c=bd'}$

3.9

$\dfrac{c = bd' = (c'a)d' = (c'd')a}{a|c}$

3.10

$$\frac{\forall a,b,c\in\mathbb{N}((a|b\ b|c)/(a|c))}{\mathrm{Trns}(|)}$$

3.11

$$\frac{\mathrm{Rflx}(R)\ \mathrm{AntSym}(R)\ \mathrm{Trns}(R)}{\mathrm{PO}(R,\mathbb{N})}$$

4.0

$x,y,z \in \mathbb{Z}$

$x^2 = x^2$

xRx

4.1

$\forall x \in \mathbb{Z}\ (xRx)$

Rflx(R)

4.2

xRy

$x^2 = y^2$

4.3

$x^2 = y^2$

$y^2 = x^2$

4.4

$y^2 = x^2$

yRx

4.5

$\forall x,y \in \mathbb{Z}\ (xRy\ /\ yRx)$

Sym(R)

4.6

$xRy\ yRz$

$x^2{=}y^2\ y^2{=}z^2$

4.7

$x^2 = y^2 = z^2$

$x^2 = z^2$

4.8

$x^2 = z^2$

xRz

4.9

$\forall x,y,z \in \mathbb{Z}((xRy\ yRz)/(xRz))$

Trns(R)

10090

4.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{Z},R)}$$

4.11

$\mathrm{EqvCl}(x,R) = \{t \in \mathbb{Z} \mid t^2 = x^2\} = \{x,-x\}$

$x=0$: $\mathrm{EqvCl}(0,R)=\{0\}$ $x\neq 0$: $\mathrm{EqvCl}(x,R)=\{x,-x\}=\mathrm{EqvCl}(-x,R)$

$\mathbb{Z}/R = \{\{0\}\} \cup \{\{-n,n\} \mid n \in \mathbb{N}, n \geq 1\}$

5.0

$(a,b),(c,d),(e,f) \in \mathbb{N}\times\mathbb{N}$

$$\frac{a+b = a+b}{(a,b)R(a,b)}$$

5.1

$$\frac{\forall(a,b)\in\mathbb{N}\times\mathbb{N}\ ((a,b)R(a,b))}{\mathrm{Rflx}(R)}$$

5.2

$$\frac{(a,b)R(c,d)}{a+b = c+d}$$

5.3

$$\frac{a+b = c+d}{c+d = a+b}$$

5.4

$$\frac{c+d = a+b}{(c,d)R(a,b)}$$

5.5

$$\frac{\forall(a,b),(c,d)\in\mathbb{N}\times\mathbb{N}((a,b)R(c,d)/(c,d)R(a,b))}{\mathrm{Sym}(R)}$$

5.6

$$\frac{(a,b)R(c,d)\ (c,d)R(e,f)}{a+b=c+d\ c+d=e+f}$$

5.7

$$\frac{a+b = c+d = e+f}{a+b = e+f}$$

5.8

$$\frac{a+b = e+f}{(a,b)R(e,f)}$$

5.9

$$\frac{\forall(a,b),(c,d),(e,f)\in\mathbb{N}\times\mathbb{N}(((a,b)R(c,d)\ (c,d)R(e,f))/(a,b)R(e,f))}{\mathrm{Trns}(R)}$$

5.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{N}\times\mathbb{N},R)}$$

5.11

$k \coloneqq a+b.\ \mathrm{EqvCl}((a,b),R) = \{(x,y)\in\mathbb{N}\times\mathbb{N} \mid x+y=k\} = \{(0,k),(1,k-1),\ldots,(k,0)\}$, *size* $k+1$

$(\mathbb{N}\times\mathbb{N})/R = \{\{(0,k),\ldots,(k,0)\} \mid k\in\mathbb{N}\}$

6.0

$(a,b),(c,d),(e,f) \in \mathbb{N}\times\mathbb{N}$

$a^2+b^2 = a^2+b^2$

$(a,b)R(a,b)$

6.1

$\forall(a,b)\in\mathbb{N}\times\mathbb{N}\ ((a,b)R(a,b))$

$\mathrm{Rflx}(R)$

6.2

$(a,b)R(c,d)$

$a^2+b^2 = c^2+d^2$

6.3

$a^2+b^2 = c^2+d^2$

$c^2+d^2 = a^2+b^2$

6.4

$c^2+d^2 = a^2+b^2$

$(c,d)R(a,b)$

6.5

$\forall(a,b),(c,d)\in\mathbb{N}\times\mathbb{N}((a,b)R(c,d)/(c,d)R(a,b))$

$\mathrm{Sym}(R)$

6.6

$(a,b)R(c,d)\ (c,d)R(e,f)$

$a^2+b^2=c^2+d^2\ c^2+d^2=e^2+f^2$

6.7

$a^2+b^2 = c^2+d^2 = e^2+f^2$

$a^2+b^2 = e^2+f^2$

6.8

$a^2+b^2 = e^2+f^2$

$(a,b)R(e,f)$

6.9

$\forall(a,b),(c,d),(e,f)\in\mathbb{N}\times\mathbb{N}(((a,b)R(c,d)\ (c,d)R(e,f))/(a,b)R(e,f))$

$\mathrm{Trns}(R)$

6.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{N}\times\mathbb{N},R)}$$

6.11

$n \coloneqq a^2+b^2.\ \mathrm{EqvCl}((a,b),R) = \{(x,y)\in\mathbb{N}\times\mathbb{N} \mid x^2+y^2=n\}$

$(\mathbb{N}\times\mathbb{N})/R = \{\{(x,y) \mid x^2+y^2=n\} \mid n\in\mathbb{N},\ \exists(x,y)(n=x^2+y^2)\}$

7.0

$$\frac{\rho(n) = [(R_0(r+s)-R_1)/2s](r-s)^n + [(R_1-R_0(r-s))/2s](r+s)^n}{\rho(n) = (-1/\sqrt{5})((1-\sqrt{5})/2)^n + (1/\sqrt{5})((1+\sqrt{5})/2)^n} \left\langle \begin{matrix} R_0 \longleftarrow 0 \\ R_1 \longleftarrow 1 \\ r \longleftarrow 1/2 \\ s \longleftarrow \sqrt{5}/2 \end{matrix} \right\rangle$$

7.1

$r-s = (1-\sqrt{5})/2,\ r+s = (1+\sqrt{5})/2$

7.2

$\rho(n) = (1/\sqrt{5})[((1+\sqrt{5})/2)^n - ((1-\sqrt{5})/2)^n]$

7.3

n=0: $(1/\sqrt{5})(1-1) = 0 = \rho(0)$ n=1: $(1/\sqrt{5})\bullet\sqrt{5} = 1 = \rho(1)$

8.0

$P(n) \coloneqq (S(2,n) = n(n+1)(2n+1)/6)$

SPf

BC

$S(2,0) = 0^2 = 0$

$P(0)$

RcS

$P(n)$

$$\begin{aligned} & S(2,n+1) \\ = \; & n(n+1)(2n+1)/6 + (n+1)^2 \\ = \; & (n+1)\bullet[n(2n+1)+6(n+1)]/6 \\ = \; & (n+1)\bullet(2n^2+7n+6)/6 \\ = \; & (n+1)(n+2)(2n+3)/6 \end{aligned}$$

$P(n+1)$

EPf

9.0

$S(1,n) = n(n+1)/2$. $P(n) := (S(3,n) = (S(1,n))^2)$

SPf

BC

$$\frac{S(3,0) = 0 = (S(1,0))^2}{P(0)}$$

RcS

$$\frac{P(n)}{\begin{array}{ll} & S(3,n+1) \\ = & (n(n+1)/2)^2 + (n+1)^3 \\ = & (n+1)^2 \bullet [n^2/4 + (n+1)] \\ = & (n+1)^2 \bullet (n+2)^2/4 \\ = & ((n+1)(n+2)/2)^2 \\ = & (S(1,n+1))^2 \end{array}}{P(n+1)}$$

EPf

10.0

$P(n) \coloneqq (\gamma(n) \geq \beta(n))$

SPf

BC

$$\frac{\gamma(0) = 0^0 = 1 = 0! = \beta(0)}{P(0)}$$

RcS

$$\frac{P(n)}{\dfrac{\beta(n+1) = (n+1)\beta(n) \leq (n+1)\gamma(n)}{\dfrac{n \leq n+1}{\gamma(n) = n^n \leq (n+1)^n}}}$$

$$\begin{aligned} & (n+1)\gamma(n) \\ = \ & (n+1)(n+1)^n \\ = \ & (n+1)^{n+1} \\ = \ & \gamma(n+1) \end{aligned}$$

$$\frac{}{P(n+1)}$$

EPf

13.0

$p,q,r \in \mathrm{Prp}$

$p \leftrightarrow p$ (*tautology*)

---

pQp

13.1

$\forall p \in \mathrm{Prp}\ (pQp)$

$\mathrm{Rflx}(Q)$

13.2

pQq

$p \leftrightarrow q$

13.3

$p \leftrightarrow q$

$q \leftrightarrow p$

13.4

$q \leftrightarrow p$

---

qQp

13.5

$\forall p,q \in \mathrm{Prp}\ (pQq/qQp)$

$\mathrm{Sym}(Q)$

13.6

$pQq\ qQr$

$p \leftrightarrow q\ q \leftrightarrow r$

13.7

$p \leftrightarrow q \leftrightarrow r$

$p \leftrightarrow r$

13.8

$p \leftrightarrow r$

---

pQr

13.9

$\forall p,q,r \in \mathrm{Prp}((pQq\ qQr)/pQr)$

$\mathrm{Trns}(Q)$

13.10

$$\frac{\mathrm{Rflx}(Q)\ \mathrm{Sym}(Q)\ \mathrm{Trns}(Q)}{\mathrm{Eqv}(Q)}$$

13.11

$\mathbb{F} \coloneqq \mathrm{Prp}/Q \coloneqq \{\mathrm{EqvCl}(p,Q) \mid p \in \mathrm{Prp}\}$

13.12

pQq, p→r, rQs given.

$$\frac{pQq}{p \leftrightarrow q}$$

13.13

$$\frac{p \leftrightarrow q}{q \rightarrow p}$$

13.14

$$\frac{rQs}{r \leftrightarrow s}$$

13.15

$$\frac{r \leftrightarrow s}{r \rightarrow s}$$

13.16

$$\frac{q \rightarrow p\ \ p \rightarrow r\ \ r \rightarrow s}{q \rightarrow s}$$

13.17

$F\mathcal{E}G \coloneqq (F \Longrightarrow G)$: *well-defined by* (ii) *(independent of representatives* $p \in F, r \in G$*)*

13.18

$$\frac{F,G,H \in \mathbb{F} \quad p,p' \in F}{pQp'}$$

13.19

$$\frac{pQp'}{p \leftrightarrow p'}$$

13.20

$$\frac{p \leftrightarrow p'}{p \rightarrow p'}$$

13.21

$$\frac{\forall F \in \mathbb{F}\ (F\mathcal{E}F)}{\mathrm{Rflx}(\mathcal{E})}$$

13.22

$$\frac{F\mathcal{E}G\ G\mathcal{E}F}{p_0 \rightarrow q_0\ q_0 \rightarrow p_0}$$

13.23

$$\frac{p_0 \rightarrow q_0\ q_0 \rightarrow p_0}{p_0 \leftrightarrow q_0}$$

13.24

$$\frac{p_0 \leftrightarrow q_0}{p_0 Q q_0}$$

13.25

$$\frac{p_0 Q q_0}{F = G}$$

13.26

$$\frac{\forall F,G \in \mathbb{F}((F\mathcal{E}G\ G\mathcal{E}F)/(F=G))}{\mathrm{AntSym}(\mathcal{E})}$$

13.27

$$\frac{F\mathcal{E}G\ G\mathcal{E}H}{\forall p{\in}F,q{\in}G(p{\to}q)\ \forall q{\in}G,r{\in}H(q{\to}r)}$$

13.28

$$\frac{\forall p{\in}F,q{\in}G(p{\to}q)\ \forall q{\in}G,r{\in}H(q{\to}r)}{\forall p{\in}F,r{\in}H(p{\to}r)}$$

13.29

$$\frac{\forall p{\in}F,r{\in}H(p{\to}r)}{F\mathcal{E}H}$$

13.30

$$\frac{\forall F,G,H{\in}\mathbb{F}((F\mathcal{E}G\ G\mathcal{E}H)/F\mathcal{E}H)}{\mathrm{Trns}(\mathcal{E})}$$

13.31

$$\frac{\mathrm{Rflx}(\mathcal{E})\ \mathrm{AntSym}(\mathcal{E})\ \mathrm{Trns}(\mathcal{E})}{\mathrm{PO}(\mathcal{E},\mathbb{F})}$$

14.0

n=3: β=216 δ=729 γ=720 α=27! (γ,δ *swapped*; *close*: 720 *vs* 729)
n=4: β=331,776 δ=4^{24}≈2.81×10^{14} γ=24!≈6.20×10^{23} α=256!
n=5: β≈2.49×10^{10} δ=5^{120}≈7.5×10^{83} γ=120!≈6.7×10^{198} α=3125!

14.1

$$\frac{331{,}776 < 4^{24} < 24! < 256!}{P(4)}$$

14.2

β<δ (*base* n=4; *other two below*): $a(n)$:=ln $\beta(n)$=n•ln(n!), $b(n)$:=ln $\delta(n)$=n!•ln n

$$\frac{a(n)<b(n)}{(n+1)\ln(n!)+(n+1)\ln(n+1) < (n+1)[n!/n\bullet\ln n+\ln(n+1)]}$$

14.3

$$\frac{n!/n \le n!-1}{a(n+1) < b(n+1)}$$

14.4

$n \le n!(n-1)$, *true* $\forall n \ge 4$ *since* $n!(n-1) \ge 24\bullet 3 = 72 \gg n$

14.5

$\delta<\gamma$: $\ln n<\ln(n!)$ $(n\geq 2)$

$\ln n < \ln(n!)$ $(n\geq 2)$

$n!\bullet\ln n < n!\bullet\ln(n!) \leq \ln((n!)!) = \ln \gamma(n)$

14.6

$\gamma<\alpha$: $n! \leq n^n$ $(n\geq 2)$, *each of* $n!$ *factors is* $\leq n$

$n! \leq n^n$ $(n\geq 2)$

$\alpha(n) = (n^n)! \geq (n!)! = \gamma(n)$

14.7

$P(4)$ 14.2 14.3 14.5 14.6

$\forall n\in\mathbb{N}(n\geq 4)$ $(\beta(n)<\delta(n)<\gamma(n)<\alpha(n))$

15.0

$Q(n) \coloneqq (\Sigma(k{=}0..n)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)] = (n{+}1)\hat{}(p{+}1))$

SPf

BC

$$\begin{aligned} & 1\hat{}(p{+}1) - 0\hat{}(p{+}1) \\ = \ & 1 \\ = \ & (0{+}1)\hat{}(p{+}1) \end{aligned}$$

$Q(0)$

RcS

$Q(n)$

$$\begin{aligned} & \Sigma(k{=}0..n{+}1)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)] \\ = \ & \Sigma(k{=}0..n)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)] + [(n{+}2)\hat{}(p{+}1){-}(n{+}1)\hat{}(p{+}1)] \\ = \ & (n{+}1)\hat{}(p{+}1) + (n{+}2)\hat{}(p{+}1) - (n{+}1)\hat{}(p{+}1) \\ = \ & (n{+}2)\hat{}(p{+}1) \\ = \ & ((n{+}1){+}1)\hat{}(p{+}1) \end{aligned}$$

$Q(n{+}1)$

EPf

15.1

$(k{+}1)\hat{}(p{+}1) - k\hat{}(p{+}1) = \Sigma(j{=}0..p)\ C(p{+}1,j)\ k\hat{}j$

$(k{+}1)\hat{}(p{+}1) - k\hat{}(p{+}1) = \Sigma(j{=}0..p)\ C(p{+}1,j)\ k\hat{}j \qquad Q(n)$

$\Sigma(j{=}0..p)\ C(p{+}1,j)\ S(j,n) = (n{+}1)\hat{}(p{+}1)$

15.2

$S(p,n) = 1/(p{+}1) \bullet [(n{+}1)\hat{}(p{+}1) - \Sigma(j{=}0..p{-}1)\ C(p{+}1,j)\ S(j,n)]$

15.3

$$\begin{aligned} & p{=}2{:}\ S(2,n) \\ = \ & \tfrac{1}{3}[(n{+}1)^3{-}(n{+}1){-}3{\bullet}n(n{+}1)/2] \\ = \ & n(n{+}1)(2n{+}1)/6 \ \checkmark \ (\textit{matches}\ 2(8)) \end{aligned}$$